

Overview

Electroencephalography is a technique that involves placing electrodes in standardised patterns across the skull and measure voltage across them. This produces a noisy output as clusters of electrodes respond to stimuli, either baseline (lack of stimuli or rest) or induced (limb movement), that becomes spread across the head through volume conduction. This project aimed to see if machine learning could correctly predict imagined left- or right-handed movement from volunteer EEG data patterns.

Dataset

EEG data was obtained OpenNeuro dataset ds004362, which was produced during a project exploring brain-computer interfaces (BCI) (1). This dataset contained recordings from 109 volunteers using 64 EEG channels positioned according to the international 10 10 system (2). The original dataset was recorded at 160Hz, whilst within the BIDS-formatted dataset volunteers 88, 92, and 100 were recorded at 128Hz.

During this study, patients were asked to imagine either left- or right-handed movement in response to a stimulus, recorded as Task 2 in the project repository

(<https://openneuro.org/datasets/ds004362/versions/1.0.0/file-display/README>). Of the 14 runs conducted on each volunteer, runs 4, 8, and 12 corresponded to task 2, approximately 45 trials per subject. Rest trials were excluded to simplify the project to a binary discrimination of left- versus right-hand discrimination. All 109 volunteers were included in the within-subject analysis, whilst cross-subject and grand average studies used the 106 participants measured at 160Hz.

Preprocessing

Bands-pass filtering of continuous EEG data between 8 and 30 Hz was performed prior to epoch extraction. These frequencies were chosen as they encompass the principal sensorimotor Mu (μ) and Beta (β) rhythms that are relevant for imagery analysis. The power-line frequency of 60Hz lays outside of this frequency, removing the need for further notch filtering. Recording-time filter settings were otherwise undocumented in the metadata of the dataset.

Classification epochs were extracted between 0.5 and 2.5 seconds following the onset from the task-cue, providing a 2 second window for analysis. There was no baseline correction, amplitude-based artefact rejection, bad-channel interpolation or independent component analysis correction, applied to these epochs.

According to the BIDS metadata, the recordings used the earlobes of patients as a reference point. It did not, however, identify which ear was used for individual participants or and there was no independent verification of the configuration across participants. As this was consistent across participants and runs it was not expected to introduce a class-specific difference but may impact the spatial mappings of scalp topographies.

For within-subject analysis, data was left at its native sampling rate. For cross-subject analysis, the 160 Hz recordings were decimated by a factor of 2, to 80Hz. To reduce the differences in signal amplitude, a participants data is scaled according to the participants signal standard deviation.

Machine Learning Approaches

Four feature-extraction and classification pipelines were investigated for how different representations of the EEG signals impacted discrimination of left- or right-handed motor imagery. A baseline was made using a simple band-power model using C3 and C4 electrodes which discriminated μ - and β -band power before using linear discrimination analysis (LDA) to classify the data. This was compared with a common spatial pattern (CSP) + LDA approach, in which weight filters were applied to variance matrices to maximise signal variance in one direction whilst reducing it in another. This data too was then fed to a LDA for classification. A tuned version of this same algorithm was then run to ensure optimal performance with the chosen retained spatial components (2, 4 or 6). The two other models trained for this study were a Dummy, designed to select the most common class and achieve accuracy close to 50%, and a Riemannian model with a linear regression classifier.

Dummy Classifier

The Dummy classifier used the “*most_frequent*” (see Github code “placeholder” for details) strategy to report the most common class (left or right) in each data set. Given the approximately 50/50 split between left- and right-handed trials, any deviation above 50% accuracy could be a sign of data leakage between sets.

Band Power + LDA

This model utilised a conventional frequency-domain representation was implemented using the C3 and C4 electrodes, located above the sensorimotor regions. Welch's method was used to estimate power spectral density. For each trial, Mu-power (8-13Hz) for C3/C4 and Beta-power (13-30Hz) were extracted, comprising 4 total components, each of which was taken to the natural log to reduce the impacts of signal magnitude. This data was subsequently fed into a LDA for classification and plotting.

This model was intended to provide a simplistic baseline focused on an oversimplified neurophysiology of motor movement. It was this model that acted as the point of comparison for the more complex models which relied on a wider spectrum of data.

CSP + LDA

Common spatial Pattern (CSP) analysis is a method that allows for identification of spatial projections that maximise the difference in signal variance between left- and right-handed trial data. CSP included all 64 EEG channels, increasing its complexity compared to Band Power + LDA, and allowing for spatial information to be extracted from across the entire scalp.

Four CSP components were retained in this model, although other models (see CSP + LDA (Tuned)) analysed if this was reducing performance. The Ledoit-Wolf regularisation method was used to transform the CSP data and improve the estimation of the covariance structure, with the logarithm of the filtered spatial components being used as the resulting feature representation. This data was subsequently into the LDA.

A single workflow was used for the CSP and LDA, with the CSP fitted to only the training data within each of the cross-validation folds. As a consequence, the spatial filters used to transform the data did so without access to test labels and data during training.

CSP + LDA (Tuned)

A second CSP + LDA model was used to ensure that the chosen number of retained characteristics, 4, was not impacting the model's outcome. Therefore, the tuned model utilised a separate cross-validation procedure to select between 2, 4, and 6 retained characteristics.

Nested cross-validation was used to keep hyperparameter selection separate from the final model evaluation, reducing the chances of data leakage. During Leave-One-Run-Out (LORO) testing, a further Leave-One-Group-Out (LOGO) step was implemented, and the recording run defining the grouping variables. The optimal number of retained CSP components was selected using the inner training data prior the refitting of the CSP + LDA data using both the training runs and the withheld data sets. Other parameters such as the range of frequencies, regularisation method or LDA configuration were tuned as a part of this model.

Riemannian + Logistic Regression

A Riemannian geometry-based approach was chosen for representation of the spatial covariance structure of each EEG trial. An Oracle Approximating Shrinkage (OAS) estimator was used to regularise the covariance matrix for each of the epochs within a data set. Since these covariance matrices were symmetric positive-definitive, they do not occupy typical Euclidian feature space, and a Riemannian approach was more appropriate, with the covariance matrices projected into Riemannian manifold space.

The tangent-space features were classified using a linear regression model, with a default configuration and a maximum of 1000 iterations for optimisation. Within each cross-validation fold, only the trial runs were used to calculate Riemannian reference mean, which was used to transform the tangent-space for classification. This model created a comparison to the CSP + LDA in which information was represented through the covariance geometry of all EEG channels rather than through chosen spatial filters.

Evaluation Strategy

Model performance was evaluated through two cross-validation strategies to differentiate between within-subject and within-volunteer EEG data analysis. The primary metric of this validation was accuracy, as the number of left- and right-handed trials was approximately even. For all trainable components within each pipeline, which includes feature transformation, spatial filters, covariance estimations, classifier parameters, and hyperparameter selection, were fitted with exclusively training data within each of the folds.

Intrasubject Evaluation

Intrasubject evaluation was completed using LORO analysis on runs 4, 8, and 12. Within each fold, the model was trained on two of the data sets, with the third being withheld and then used as testing data. This process was performed a total of 3 times, so that each of the runs could serve as testing data, and the mean accuracy taken from these values.

This strategy was employed to assess if EEG datasets contained sufficient features to distinguish left- from right-handed motor imagery in the context of unseen participant runs. Whilst the individuals set contributed to both the learning and testing data, no individual recording was present in both sets at the same time. This analysis is more stringent than random partitioning of a volunteers runs, which may include the same runs in both teaching and testing data.

Whilst LORO analysis was used the principal method for intrasubject evaluation, an earlier shuffled 5-fold cross-validation was used for descriptive analysis for comparison of other more stringent approaches.

Intersubject Evaluation

Ten-fold GroupKFold was used to separate the cohort into teaching and testing cohorts, assessing whether analysis from the models generalised to participants withheld from the testing data. Each participants identity was used as a grouping variable to ensure that all their runs were held out in a single group, ensuring there is no leakage between teaching and testing groups. The intrasubject evaluation included 106 participants recorded at 160Hz, which was reduced to a common 80Hz sampling rate following decimation.

Accuracy at the level of each participant was calculated from out-of-fold predictions, creating accuracy calculations for each volunteer, which were subsequently used for statistical analysis (see Statistical Analysis section).

Statistical Analysis

Permutation Testing of CSP + LDA

A permutation test was used to assess if the CSP + LDA was able to classify a participant's data at a level greater than that of chance, thus rejecting or accepting the null hypothesis. This test permuted each participant's class label 200 times prior to the model being refitted with each of these permutations, and the model run again.

The accuracy of this permuted model was compared to the correctly labelled data using an upper-tail test, with the permuted (p)-value being correcting with:

$$p = \frac{1 + C}{1 + N}$$

where “C” is the number of participants whose data was greater than or equal to the observed accuracy, and “N” is the total number of permutations (N=200).

False Discovery Rate Correction

Due to the permutation rates being calculated independently from the 109 participants, a Benjamin-Hochberg false discovery rate was applied ($q = 0.05$) to the permutation (p)-values. This correction allows for control of the proportion of false positives in the set of statistically significant participants. For the CSP + LDA model, the 35 significant participants were reduced to 24 following this correction. By applying this measure, the consistency of above-average participants could be assessed instead of relying on exclusively aggregates of participant accuracy.

Intrasubject Model Comparison

A paired Wilcoxon signed-rank test was used to compare results between models, an appropriate measure given that all participants were used in all models, creating a paired observation. The CSP + LDA models were compared to the other 4 models (see Machine Learning Approaches), and distribution of paired participant-level accuracies that were systematically displaced from 0 was assessed. Two-sided tests were applied to these measures with a threshold of $p = 0.05$.

Intersubject Model Comparison

For each of the models, a one-sample Wilcoxon signed-rank test was used to measure if participant accuracy differed from the 0.5 chance level expected for random classification of data classes. In addition to this, the CSP + LDA and Riemannian model were compared directly with a paired Wilcoxon signed-rank test to assess if the paired difference between these models for intrasubject accuracy was significantly different from zero.

Results

Intrasubject Classification Performance

LORO cross-validation was used across all 109 subjects (Figure 1), with mean accuracy and varying substantially (Table 1) between the approaches explored.

The Dummy model performed closely to the expected and desired outcome of 0.5 chance level, scoring a mean accuracy of 0.501 and median of 0.511.

The Band Power + LDA model improved on this performance with a mean accuracy of 0.557 and a median of 0.556. This approach also found that 36 of the subjects exceeded the cut-off for random chance of 0.6.

CSP + LDA once again improved on this score, with a mean accuracy of 0.602 and a median of 0.578, and 49 subject over the 0.6 threshold. Compared to this, the tuned CSP + LDA model had a mean of 0.603, and a median of 0.564, with 48 subjects over the threshold.

Riemannian + LR has the highest overall performance for intrasubject validation, with a mean of 0.637, median of 0.622, and 70 participants exceeding the 0.6 chance level threshold.

Figure 1 – LORO Accuracy by Model

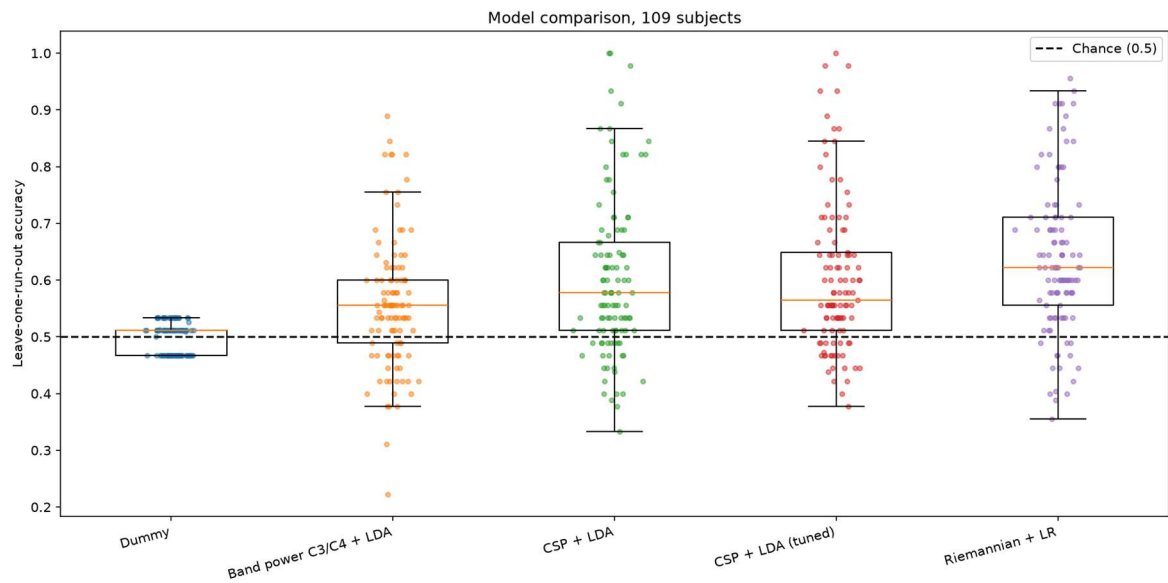


Table 1 – Descriptive Statistics for LORO Model Output

Model	Mean Accuracy	Median Accuracy	Subjects ≥ 0.60
Dummy	0.501	0.511	0
BP C3/C4 + LDA	0.557	0.557	36
CSP + LDA	0.602	0.578	49
CSP + LDA (Tuned)	0.603	0.564	48
Riemannian + LR	0.637	0.622	70

Statistical Analysis of Intrasubject Classification Models

Paired Wilcoxon signed-rank tests were used for the comparison of all models to the chosen CSP + LDA model (Table 2).

The CSP + LDA model outperformed both the Dummy model and Band Power + LDA model, with a mean paired difference of +0.101 ($p = <0.0001$) and +0.044 ($p = 0.0016$) respectively.

There was no statistically significant difference between the CSP + LDA and the tuned variant of the same model, with the mean paired difference being -0.001 ($p = 0.9855$).

Riemannian + LR outperformed the CSP + LDA model with a mean difference of -0.035 (from the CSP + LDA perspective) ($p = 0.0001$).

Table 2 – Paired Comparison: CSP + LDA vs Other Models (Paired Wilcoxon Signed-Rank Test)

Model	Mean Difference	p
Dummy	+0.101	0.0000
BP C3/C4 + LDA	+0.044	0.0016
CSP + LDA	-0.001	0.9855
Riemannian + LR	-0.035	0.0001

Intersubject Classification Performance

The intersubject classification was evaluated using 106 participants, with models being tested on datasets they did not see during training.

The Dummy classifier achieved a pooled accuracy of 0.505, a mean accuracy per subject of 0.505, and a median of 0.511. Its performance did not differ significantly from chance ($p = 0.8340$).

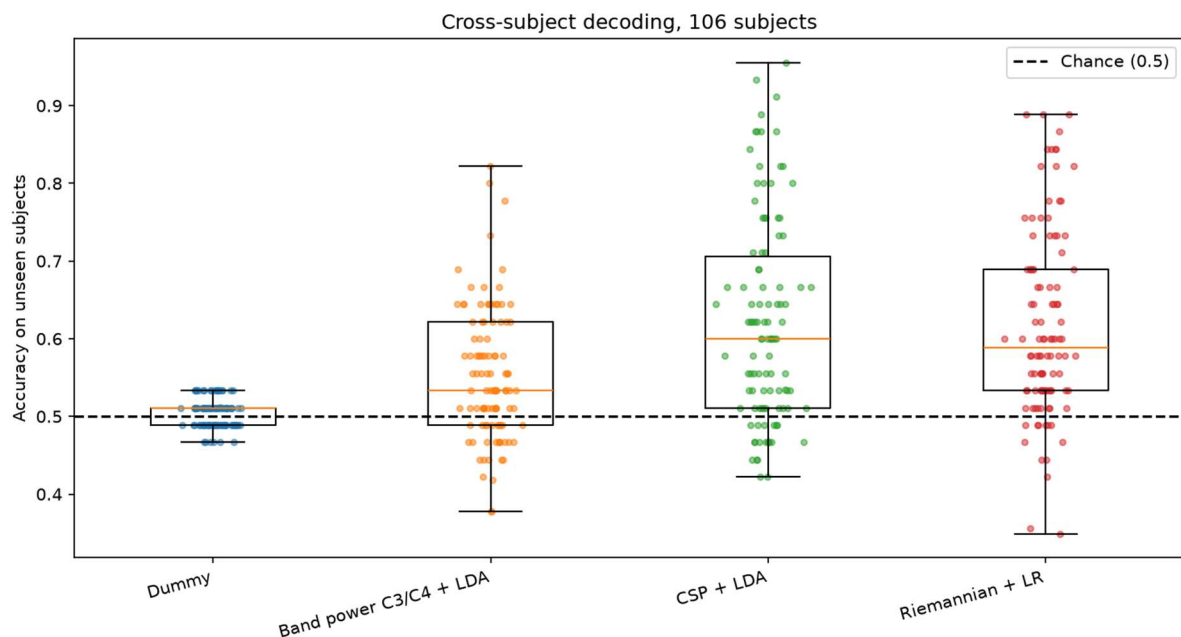
Band Power + LDA achieved a pooled accuracy of 0.553, mean accuracy per subject of 0.553, and a median of 0.533. It differed significantly from chance ($p = <0.0001$).

CSP + LDA achieved a pooled accuracy of 0.621, a mean accuracy per subject of 0.621, and a median of 0.600. It differed significantly from chance ($p = <0.0001$).

Riemannian + LR achieved a pooled accuracy of 0.616, a mean accuracy per subject of 0.616, and a median of 0.589. It differed significantly from chance ($p = <0.0001$).

The mean difference between the CSP + LDA and Riemannian + LR models was +0.005, however this result was not statistically significant ($p = 0.6113$). This indicates no statistical advantage of one model over the other when generalised to intersubject analysis.

Figure 2 – Cross Subject Accuracy by Model



Intrasubject Versus Intersubject Generalisation

The difference between these two methods of producing test versus teaching data were analysed through comparison of mean accuracies from the two approaches.

The Dummy classifier showed a minimal change (drop = -0.004) between LORO (0.500) and Cross-subject analysis (0.505).

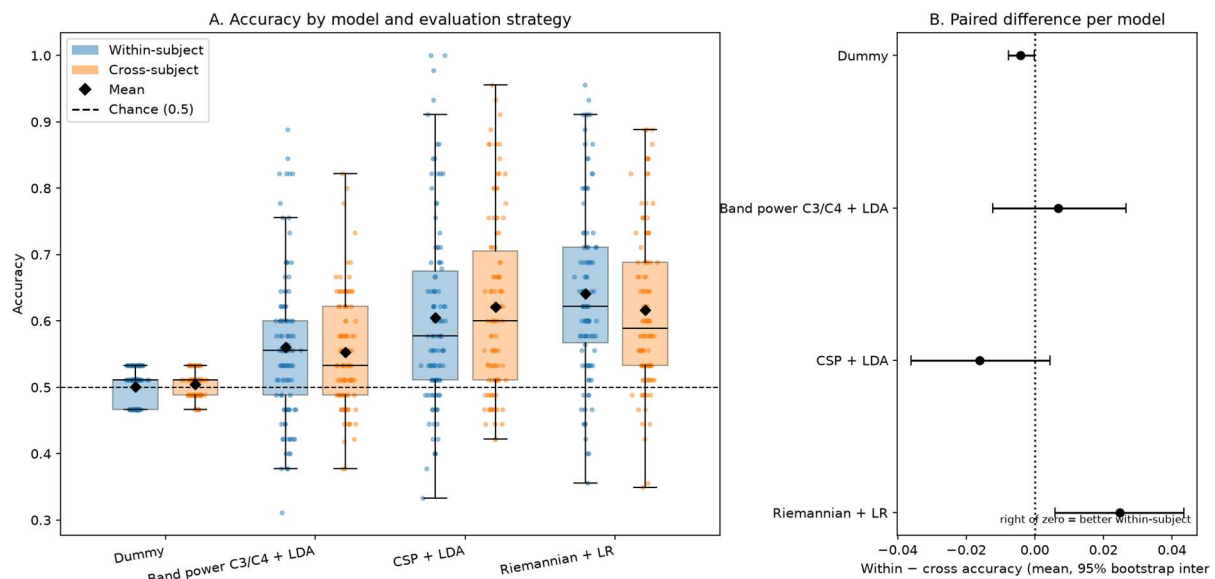
Band Power + LDA demonstrated a slight drop (drop = +0.007) from LORO (0.560) to Cross-subject analysis (0.553).

CSP + LDA showed a slight increase (drop = -0.016) between LORO (0.605) and Cross-subject analysis (0.621).

Riemannian + LR showed the largest change (drop = +0.025) from LORO (0.641) and Cross-subject (0.616).

This comparison indicates the ability of the CSP + LDA and Riemannian + LR approach to generalise well between unseen runs and unseen participants. However, Riemannian + LR demonstrated a modest reduction in mean average between the two approaches.

Figure 3 – A. Accuracy by Model and Evaluation Strategy; B. Paired Difference per Model



Time-Frequency Analysis

Grand-average time-frequency analysis (TFA) was performed across 106 subjects (see Dataset for exclusion of 3 participants) to explore oscillatory power between left- and right-handed movement. Analysis focused on the Mu (8-13Hz) and Beta (13-30Hz) frequency ranges across electrodes C3 and C4, which overlay the primary motor cortex. The mean left- and right-handed difference was calculated for epochs for each participant across the 0.5-2.5 second measurement intervals.

The resulting curves demonstrate opposing pattern at both C3 and C4, particularly in the Beta frequency. At C3, the mean Left-Right difference was positive across both Mu and Beta frequencies, with mean differences of +0.0186 and +0.0154 log ratio units respectively. The effect of C3 Mu reached nominal statistical significance ($t = +1.99$, $p = 0.0490$) while the C3 Beta effect was stronger ($t = +2.99$, $p = 0.0035$).

At C4, the corresponding differences had negative results, with mean values of -0.0155 for Mu frequencies and -0.0212 for Beta frequencies. The C4 Mu effect was not statistically significant ($t = -1.41$, $p = 0.1622$), whilst the C4 Beta effect was statistically significant ($t = -3.96$, $p = 0.0001$).

The results generated were based off 4 exploratory, uncorrected statistical analysis. Using a Bonferroni-adjusted significance threshold of 0.0125, the C3 Beta and C4 Beta effects remained

significant, whilst the C3 Mu effect did not. Therefore, the most robust time-frequency evidence was observed in the Beta frequencies, with positive Left-Right differences at C3 and negative differences at C4.

Figure 4 – Time-Frequency Left – Right Differences

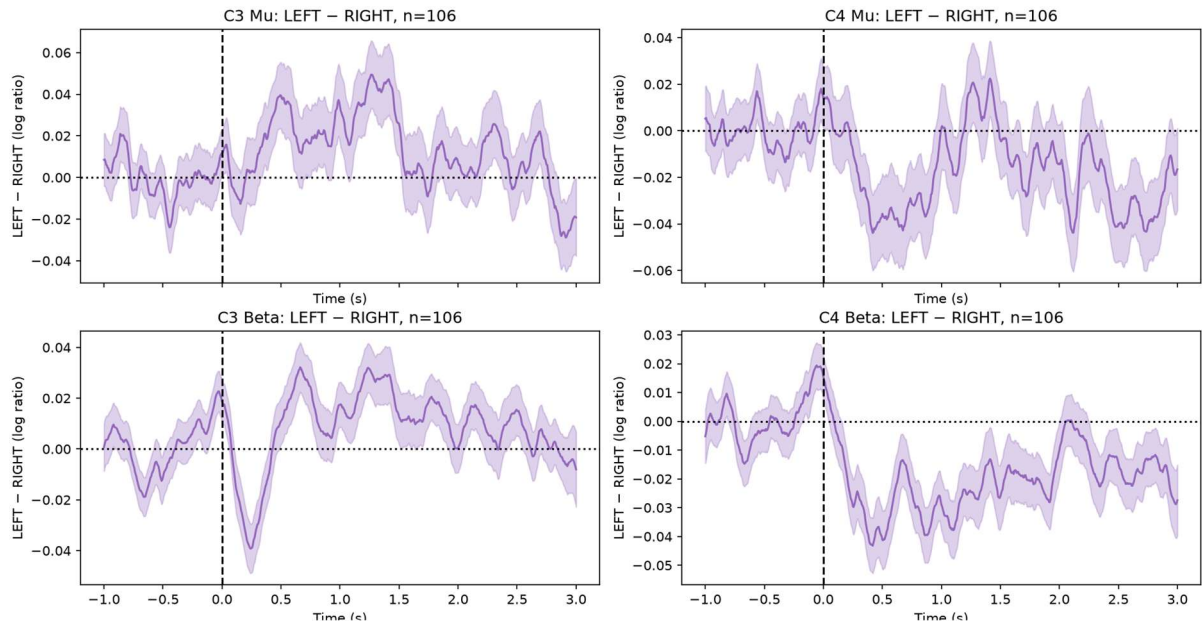
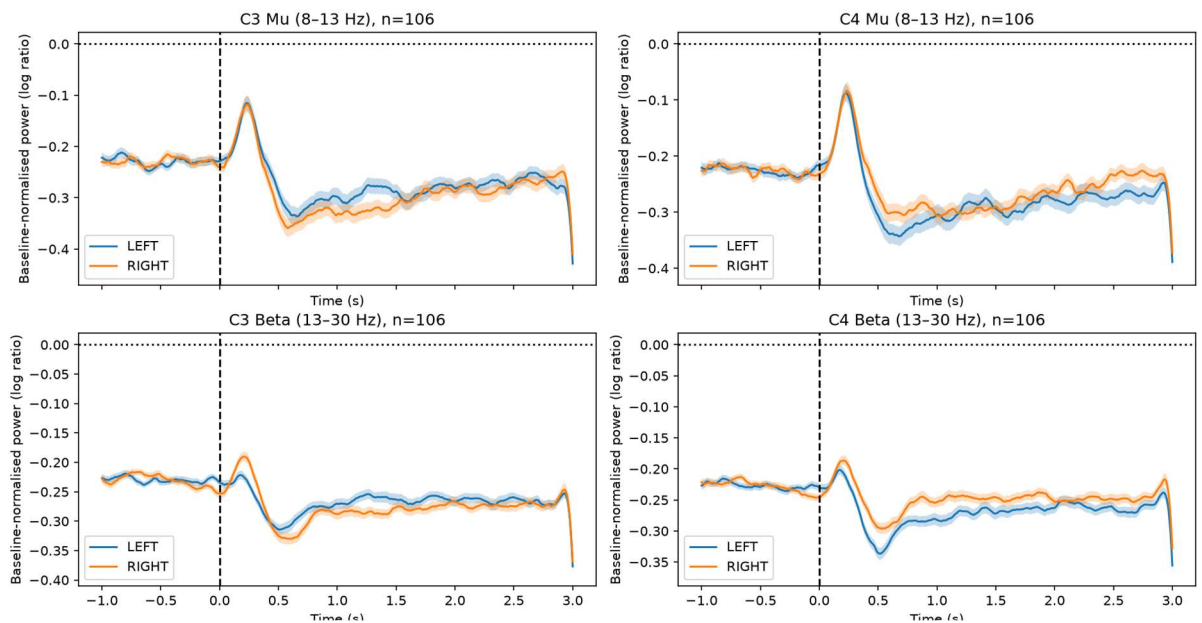


Figure 5 – Condition Specific C3/C4 Power



CSP Spatial Patterns

CSP spatial patterns were explored to examine electrode distributions contributing to discrimination of left- and right-handed sensorimotor EEG data. Four CSP components were retained from CSP +

LDA analysis, with the resultant pooled data demonstrating clusters of activity across the sensorimotor and frontocentral electrode regions when plotted on a scalp map (Figure....).

Component 1 demonstrated greater mean component power during right-handed trials, with the strongest spatial weights appearing at Cp4, Cp6, C4, P4, and P6. Contrasting this, component 2 showed greater mean component power during left-handed trials, with the weights at Cp3, C3, Cp5, Cp1, and P3.

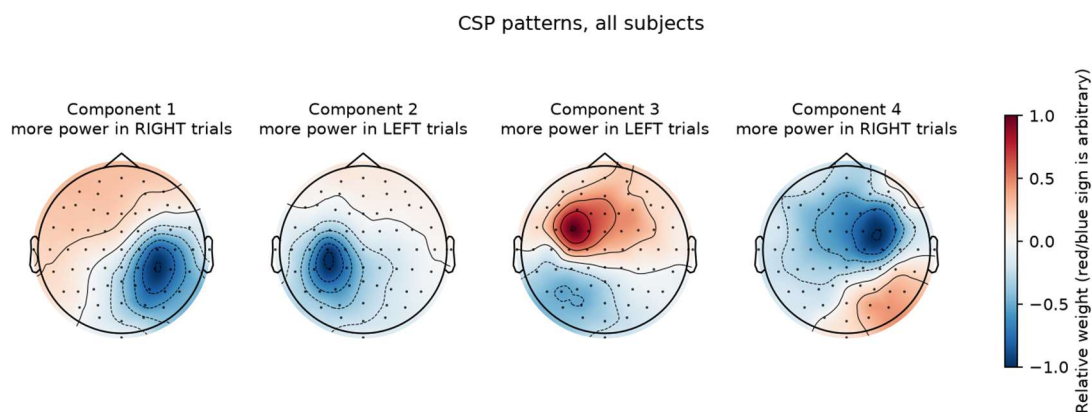
Components 3 and 4 appeared with greater emphasis in the frontotemporal regions of the brain. Component 3 had greater mean component power during left-handed trials, with the strongest spatial weights at Fc3, Fc1, F3, F1, and Fz. Component 4 had greater mean component power during right-handed trials, with the strongest spatial weights at Fc3, Fc2, C2, C4, and F4.

The means powers of these components support the finding of class-specific differences. Component 1 showed a mean log-power of -0.856 for right trials versus -1.087 for left trials, corresponding to a right-greater difference of -0.231 when calculated as left minus right. Component 2 showed greater left-trial power (-0.881 versus -1.078) corresponding to a left-greater difference of +0.196.

Component 3 similarly demonstrated left-greater difference (-0.863 versus -1.009), with a difference of +0.147. However, component 4 showed a right-greater difference of -0.138 (-0.985 versus -0.847).

Therefore, the CSP pattern demonstrated spatially distinct representations associated with bilateral involvement of the sensorimotor and frontocentral regions. These maps plot discriminative patterns learned from CSP rather than measures of local activation or anatomical source localisation. Notably, these maps should not be interpreted as activation versus inhibition.

Figure 6- Scalp Maps of CSP Learned Spatial Patterns



Discussion

Maini Findings

This investigation explored a machine learning approach the discrimination and classification of two movement classes (imagining left- and right-handed) input from a 64-lead EEG dataset, with emphasis on weather spatial and covariance-based representations could outperform simplistic baseline models. This study found that incorporating spatial information substantially improved the performance of models, with CSP and Riemannian approaches outperforming baseline models.

The strongest performance within-subject was the Riemannian approach paired with logistic regression, with a mean accuracy of 0.637. CSP + LDA demonstrated a mean of 0.603, outperforming both the Dummy model (0.637 vs 0.503) and Band Power at C3/C4 + LDA (0.637 vs 0.557).

In contract, when analysed with cross-subject methods, CSP + LDA produced the greatest accuracy at 0.621, versus 0.616 for the Riemannian + LR model, however, this difference was not statistically significant ($p = 0.6113$). The similar performance of both models suggest that both can capture EEG characteristics that generalise across volunteers, despite different approaches to data representation.

Impact of Spatial Data on Model Output

The performance of the Band Power at C3/C4 + LDA model in comparison to the CSP + LDA model suggests that the performance of a given model is improved when not limited to only four data components (C3 Mu, C3 Beta, C4 Mu, and C4 Beta). When incorporating all 64 EEG channels, the CSP model can provide weighted matrices that maximise differences between channels, producing spatial projections of the data. This approach is supported by the increase in mean from the Band Power model (mean accuracy = 0.557) to the CSP + LDA model (0.637), and suggests that inclusion of multichannel spatial information improved the performance of the models.

The scalp maps generated from CSP + LDA projections support this finding, with the strongest weights for the first two components clustering around the C3 and C4 electrodes. However, upon inclusion of the remaining EEG channels, a second cluster around the frontocentral region becomes apparent, indicating the model utilised patterns beyond those of the “logical” motor region.

An important distinction, however, is that the scalp maps do not correspond to a representation of contralateral control of motor function, or any other anatomical localisation of neurone activity. The scalp maps are spatial representations of CSP patterns, the output of which is impacted by volume conduction and recording reference. Thusly, the value of this experiment is in the ability of CSP to exploit spatial patterns to differentiate between motor control.

Within-subject and Cross-subject Generalisation

A finding from this study was the differences in model outputs when assessed with data from withheld runs and withheld participants. Within-subject, the Riemannian model scored the highest accuracy, 0.637, but decreased to 0.616 when tested with cross-subject data. The opposite effect was seen in the CSP model, with its accuracy rising from 0.602 for within-subject analysis up to 0.621 for cross-subject analysis.

This small change between the accuracy output of the model for within-subject versus cross-subject suggests that the model was able to capture enough spatial information from the dataset to generalise between participants. This is a desirable trait for biotechnology such as Brain-Computer Interfaces, where having to calibrate the machine for every new user is impractical.

The Riemannian model also generalised well between the two data approaches. Irrespective of its reduction in cross-subject testing, it remained well above the expected threshold for chance and was statistically comparable to the CSP + LDA model.

Interpretation of Time-Frequency Results

The time-frequency analysis served as complementary evidence to the existence of sufficient information with the EEG to support discrimination of left- or right-handed movement. The observation of an inverse relationship between the C3 beta (+0.0154 LRUs) and C4 beta (-0.0212 LRUs) during the task interval suggests a pattern of differing band power in response to stimuli. This pattern is consistent with contralateral control of motor movement. The C3 Mu effect reached nominal significance ($p = 0.0490$) but did not survive the Bonferroni correction, whilst the C4 Mu effect was not statistically significant. This suggests that the beta band provides stronger statistical evidence for the differentiation of left- and right-handed motor imagery.

CSP Tuning and Model Complexity

Tuning the CSP model to explore varying numbers of retained components did not improve its performance over the untuned CSP model, with tuned accuracy reaching 0.603 versus 0.602 for the CSP model. This increase was found to not be significant in the paired model comparison tests ($p = 0.9855$). Notably this is not evidence that hyperparameter tuning is unnecessary in all contexts, with other model parameters such as frequency range and regularisation strategy not being explored.

Complexity of the models also has a direct impact on the outcome of the model's performance, and its success in identifying discriminative information. The increase in model accuracy between the Band Power at C3 and C4 model (mean accuracy = 0.557), CSP model (mean accuracy = 0.602), and Riemannian model (mean accuracy = 0.637) supports the claim that capturing spatial information from all 64 bands increases model accuracy. This helps to justify the increased complexity of these models.

Furthermore, results for the CSP and Riemannian models in the context of between- versus cross-subject suggests a trade-off between generalisation and patient-specific outputs. In the case where it is preferable to have the highest accuracy on individual subjects, the Riemannian model performs better (mean accuracy = 0.637 versus 0.602), however, when generalisation to large cohorts is preferable, the CSP model performs better (mean accuracy = 0.621 versus 0.616).

Limitations

Several limitations within this project need to be recognised prior to interpretation. Firstly, the preprocessing approach was relatively fixed and no optimisation was performed. The EEG was band-pass filtered from 8-30Hz to represent physiological knowledge of band frequencies involved in motor control, however as this frequency was not optimised, pertinent signals may have been overlooked.

Secondly, this approach did not include any ICA-based removal of artefacts, bad-channel interpolation, or additional artefact rejection. This was done in order to increase the ease and reproducibility of the pipeline; however, it means that residual measurement of physiological artefacts could skew datasets.

Thirdly, there were differences in the EEG measurement frequencies between some of the participants. Whilst the dataset specifies that measurement was done at 160Hz, interrogation of the

data revealed that 3 of the participants had been measured at 128Hz, this may be because of an error in the code. Another issue found in the metadata of the study was inconsistent referencing points.

Whilst the earlobe was used for the reference of EEG measurement, specific details of which lobe could not be found in the metadata of the dataset. Using a single earlobe can bias results to a single hemisphere, this however, is preventable using linked earlobe references (A1/A2).

Another limitation is the use of pooled CSP spatial outputs for the generation of the scalp maps, meaning they should be interpreted as descriptive representations of spatial structure and not evidence of predictive generalisation. The supporting evidence for these learned spatial structures, the time-frequency analysis, should also be interpreted with some care, given that the C3 Mu result failed to survive the Bonferroni-adjusted threshold.

Future Works

Possibly the most consequential next step would be validation of the models on completely unseen datasets, exploring whether the models are able to generalise to participants and experimental conditions separate of those seen in this dataset.

Several improvements to the approach could be undertaken prior to this verification, namely systematic optimisation of the frequency range, epoch duration, CSP regularisation, and parameters for the classifier. Artefact correction and bad-channel rejection could be introduced to investigate if signal-quality impacts the output of the models.